31. MEP GASTROINTESTINAL TRACT 1 : DELIMITATION OF THE EMBRYO

DURATION : 05:35

STUDY SHEET

COURSE SUMMARY

This video delves deeply into the establishment of the gastrointestinal tract and the delimitation of the embryo, focusing on the development of the stomach, duodenum, and colon. Key concepts such as developmental synchronicity, the dynamics of three-dimensional movements, and the importance of the diaphragm for delimitation are explored. The relationships between different organs, the peritoneum, and associated structures are detailed, including elements like the mesogastrium and omenta. This content provides a solid foundation for any therapeutic practice, emphasising the importance of a comprehensive understanding of embryological processes in osteopathy.

ESTABLISHMENT OF THE GASTROINTESTINAL TRACT AND DELIMITATION OF THE EMBRYO

We will study the **establishment** of the **gastrointestinal tract**, the **peritoneal cavity**, and the **peritoneum** in the development of the stomach, duodenum, small intestine, and colon.

Let's begin by understanding the development of the stomach in relation to the **liver**, **spleen**, and **pancreas**. It is essential to grasp the initial phase, which is called the **delimitation of the embryo**. We have observed the integration of the **external coelom** into an **internal coelom**. At this stage, we can see the integration of the peritoneal cavity, both in the sagittal and transverse planes.

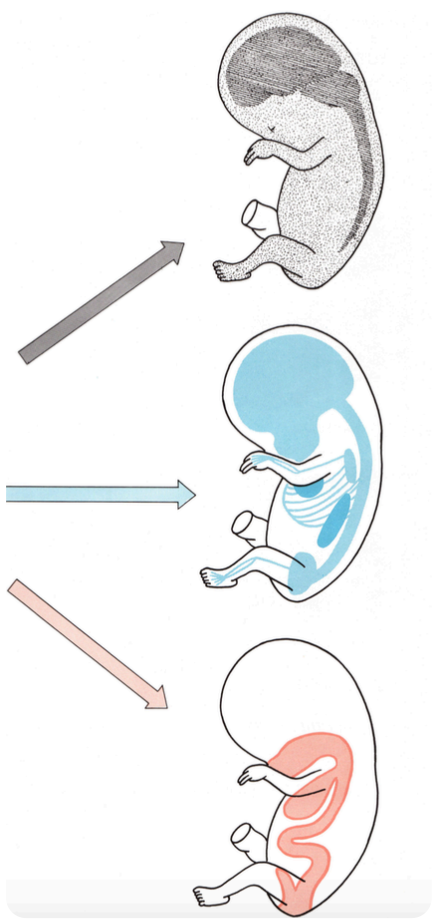

FIG. — Embryonic germ layers

Do not forget the **synchronicity** of development. Growth is the great driver of this process, with the formation of our body in a primitive order. The overall growth of the embryo, its coiling, and its return to the midline are three-dimensional movements, and even in the **fifth dimension**. The fourth dimension represents time, while the fifth dimension refers to the **energetic** and **electrical** component that underlies the entire system.

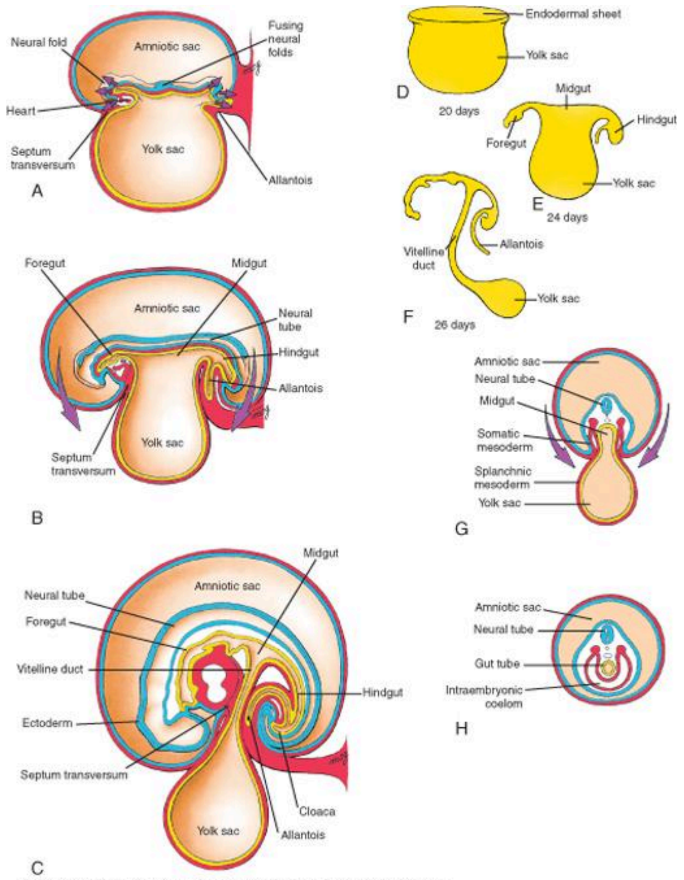

FIG. — Lateral embryonic folding

This movement is always integrated into the phase of **ectoderm ascent** and **relative endoderm descent**, with a phase almost at the level of the heart. The ascent of the ectoderm and the descent of the endoderm, as well as the integration of the heart, stretch the **primordia** of the diaphragmatic musculature and the future anterior neck muscles. To treat and reorganise this space, it is crucial to understand that the **diaphragm** must be treated and released, as it is what creates this global delimitation.

The ascent of the **neural tube**, the development of the peritoneal cavity, and the establishment of the heart form a movement that globally reintegrates towards a fulcrum called the **umbilical cord**. This cord contains the history of this development, this delimitation, and this reorganisation of the peritoneum.

This process is integrated into specific **rotational movements**, organised in clockwise and anti-clockwise directions. On a frontal plane, we observe an overall **anti-clockwise** developmental movement, while on a transverse plane, the movement is **clockwise**. The peritoneal cavity begins to organise and delimit itself, which implies that it will move.

The peritoneum surrounds an organ, which is called a **visceral peritoneum**. When a peritoneum connects an organ to the posterior part, it is called a **mesentery**. For example, for the stomach, it is called a **mesogastrium**, for the duodenum, a **mesoduodenum**, and for the colon, a

mesocolon. Depending on the location, it can also be referred to as a **fascia of Tholt**. For the small intestine, it is called a **mesentery**.

The peritoneum at the posterior part is called the **posterior parietal peritoneum**, while at the lateral part, it is referred to as the **anterior parietal peritoneum**. If a peritoneum connects one organ to another organ, it is called an **omentum**. For example, the lesser omentum connects the liver to the stomach, and there is also an omentum connecting the spleen to the stomach.

An omentum or a mesentery is always a double layer that contains a vessel in the middle. In developmental movement, folds and reorganisation occur, allowing the omentum, for example the mesentery, to move to the side. This forms a **fusion fascia**, which can be resected by surgeons without problems, as there is no vessel. A mesentery, on the other hand, always contains a vessel, while a ligament or a fascia does not necessarily contain a vessel.

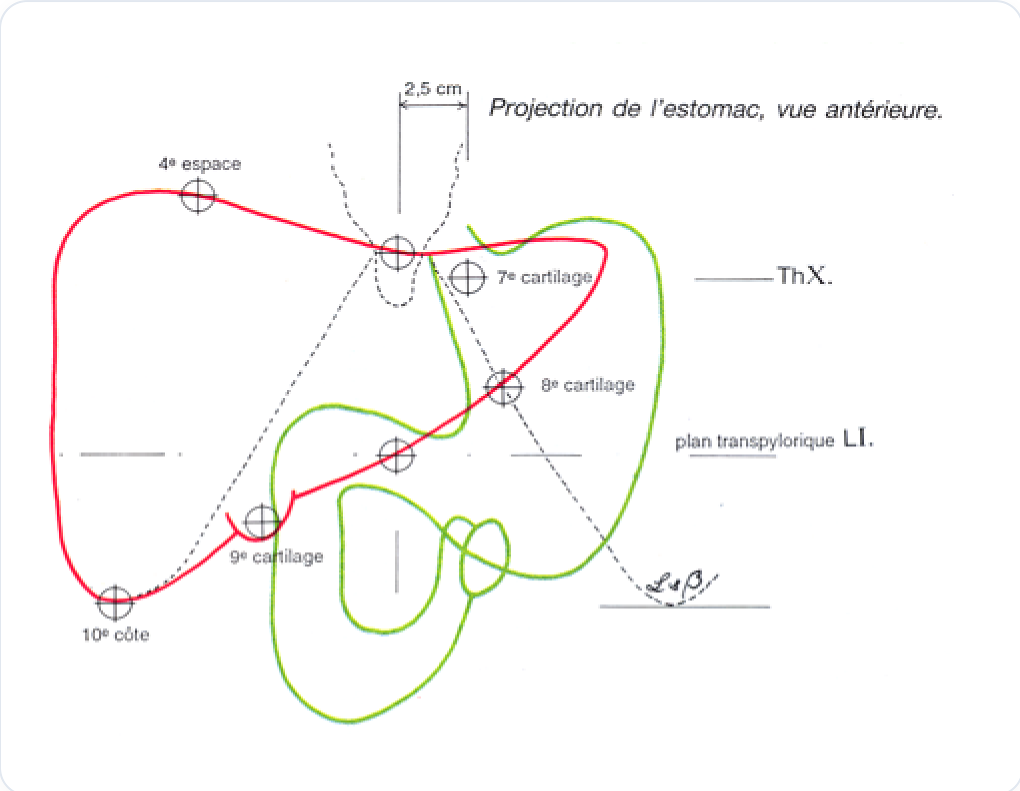

FIG. — Anterior stomach projection

At the level of the peritoneum, we have the **mesoderm**, the **somatopleure**, the **splanchnopleure**, as well as the ventral and dorsal **mesogastrium**. We also have mesenteries, fasciae, **omenta** (epiploons), and ligaments. All of this constitutes the peritoneum, organised within the peritoneal cavity.

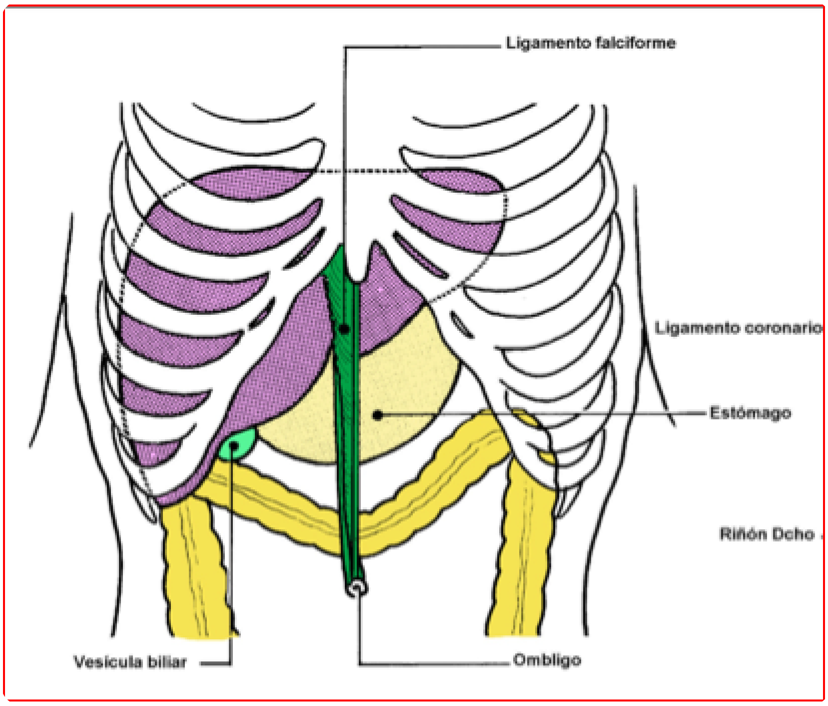

FIG. — Upper abdominal anatomy

The intestine begins to grow, the liver develops, the stomach wants to express itself, and the lungs push. All of this must find its place.

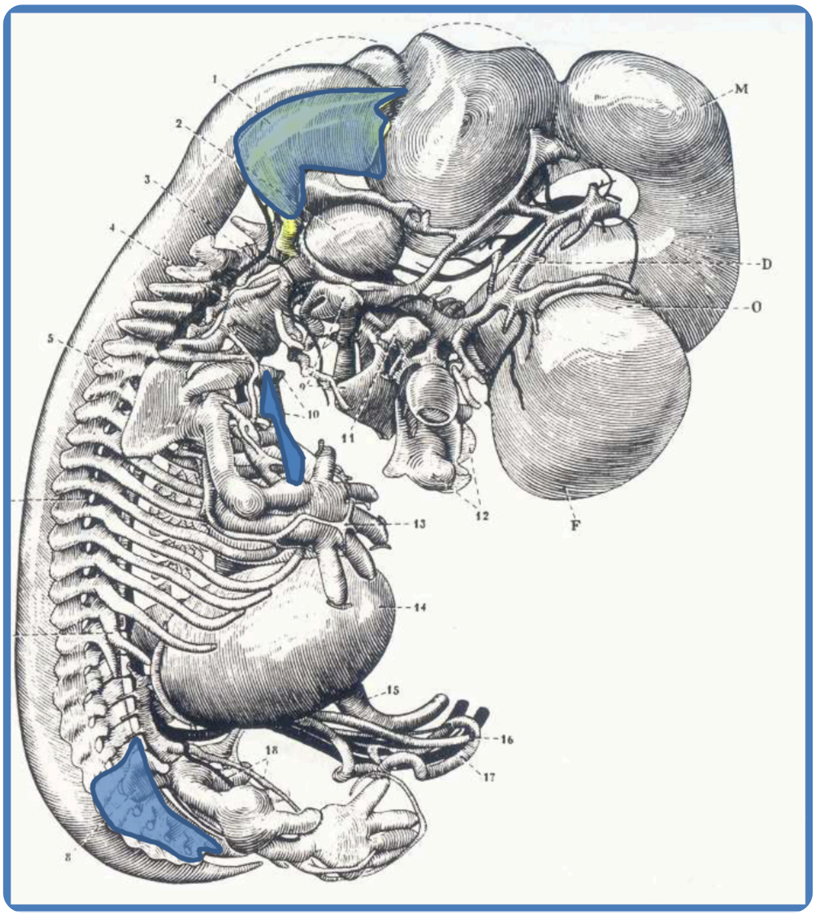

FIG. — *Human embryo, organs*

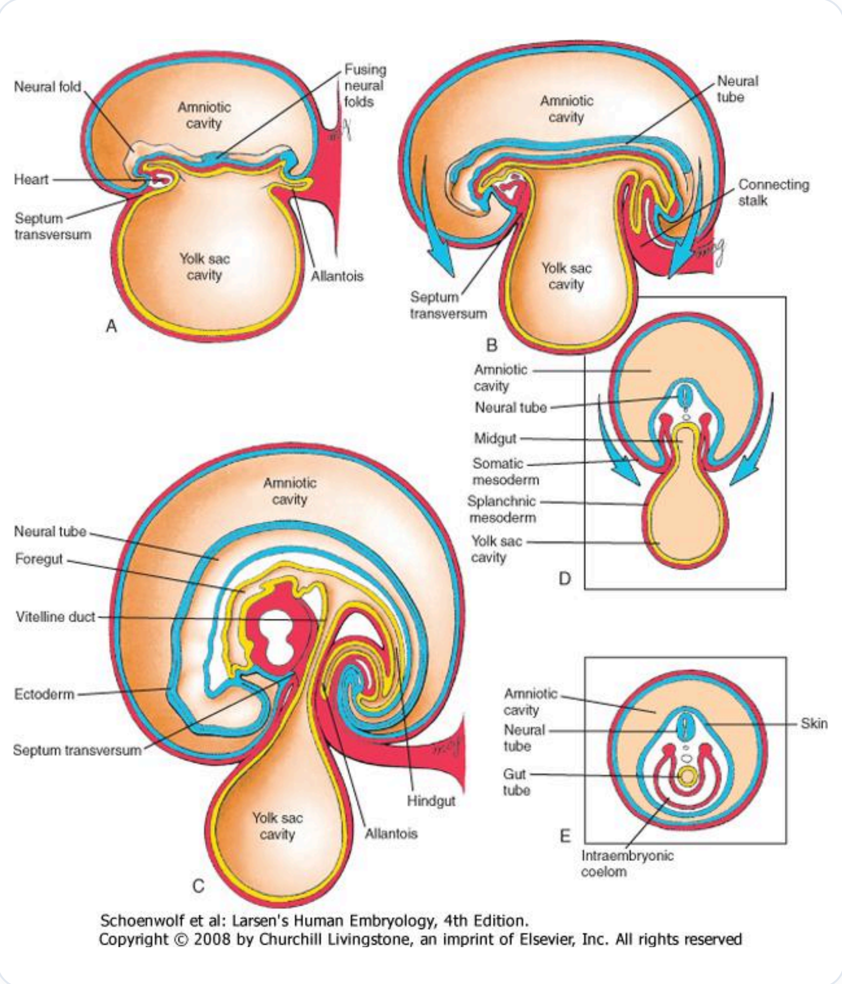

FIG. — Lateral/craniocaudal embryonic folding

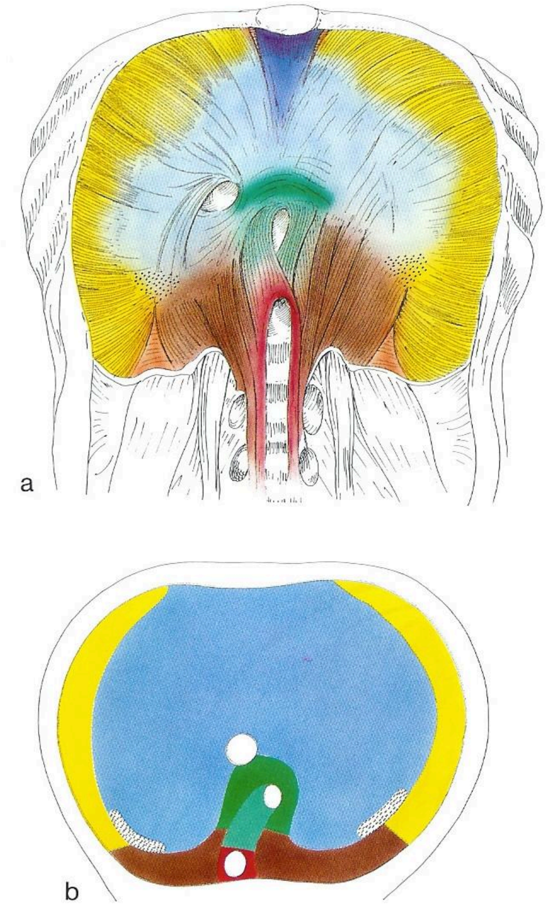

FIG. — *Human diaphragm*

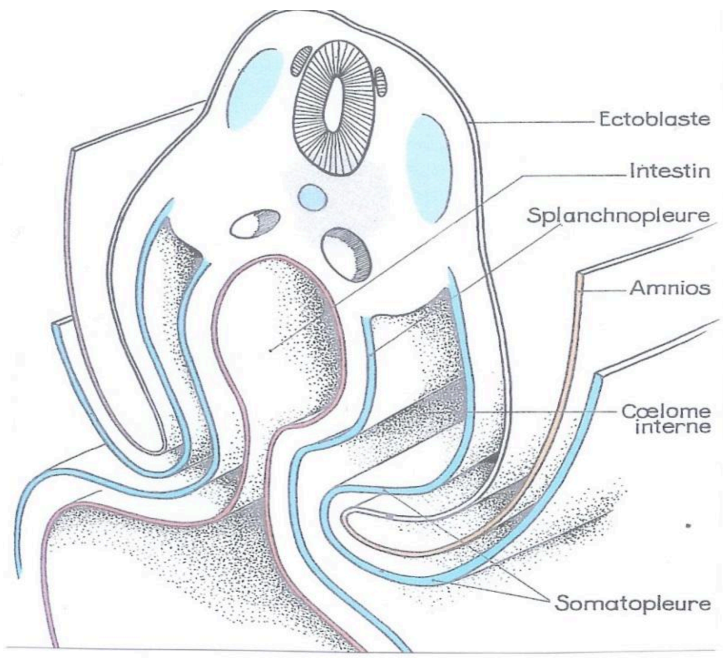

FIG. — Lateral embryonic folding